2) Numeric example (complete step-by-step)

Dataset (2D points):

- A = (1,2)
- B = (1,4)
- C = (3,2)
- D = (5,8)
- E = (6,8)

Let $K = 2$. Start with initial centroids:

- $\mu_1 = (1, 3)$
- $\mu_2 = (5.5, 8)$

We compute squared Euclidean distance $\|x - \mu\|^2 = (x_1 - \mu_1)^2 + (x_2 - \mu_2)^2$ for each point to each centroid.

Distances to $\mu_1 = (1, 3)$:

- A: $(1 - 1)^2 + (2 - 3)^2 = 0 + 1 = 1$
- B: $(1 - 1)^2 + (4 - 3)^2 = 0 + 1 = 1$
- C: $(3 - 1)^2 + (2 - 3)^2 = 4 + 1 = 5$
- D: $(5 - 1)^2 + (8 - 3)^2 = 16 + 25 = 41$
- E: $(6 - 1)^2 + (8 - 3)^2 = 25 + 25 = 50$

Distances to $\mu_2 = (5.5, 8)$:

- A: $(1 - 5.5)^2 + (2 - 8)^2 = (-4.5)^2 + (-6)^2 = 20.25 + 36 = 56.25$
- B: $(1 - 5.5)^2 + (4 - 8)^2 = 20.25 + 16 = 36.25$
- C: $(3 - 5.5)^2 + (2 - 8)^2 = (-2.5)^2 + (-6)^2 = 6.25 + 36 = 42.25$
- D: $(5 - 5.5)^2 + (8 - 8)^2 = (-0.5)^2 + 0 = 0.25$

- E: $(6-5.5)^2+(8-8)^2=(0.5)^2+0=0.25$

Now assign each point to nearest centroid (compare squared distances):

- A: 1 vs 56.25 → **Cluster 1**
- B: 1 vs 36.25 → **Cluster 1**
- C: 5 vs 42.25 → **Cluster 1**
- D: 41 vs 0.25 → **Cluster 2**
- E: 50 vs 0.25 → **Cluster 2**

Compute within-cluster SSE (sum of squared distances to assigned centroid):

- Cluster 1 (A,B,C): $1+1+5=7$
- Cluster 2 (D,E): $0.25+0.25=0.5$

Total SSE (initial) = $7+0.5=7.5$.

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3) Recompute centroids (means) and show SSE after update

New centroids (means of points assigned):

- $\mu_1^{\text{new}}=\text{mean}(A,B,C)=\left(\frac{1+1+3}{3},\frac{2+4+2}{3}\right)=\left(\frac{5}{3},\frac{8}{3}\right)\approx(1.6667,\ 2.6667)$.
- $\mu_2^{\text{new}}=\text{mean}(D,E)=\left(\frac{5+6}{2},\frac{8+8}{2}\right)=(5.5,\ 8.0)$ — note μ_2 stays the same.

Compute squared distances to new centroids and SSE:

Distances to $\mu_1^{\text{new}}=(1.6667,2.6667)$:

- A: $(1-1.6667)^2+(2-2.6667)^2\approx0.4444+0.4444=0.8889$
- B: $(1-1.6667)^2+(4-2.6667)^2\approx0.4444+1.7778=2.2222$
- C: $(3-1.6667)^2+(2-2.6667)^2\approx1.7778+0.4444=2.2222$

Cluster 1 SSE ≈ $0.8889+2.2222+2.2222=5.3333$.

Cluster 2 SSE (unchanged centroids) = $0.25+0.25=0.5$.

Total SSE after update ≈ $5.3333+0.5=5.8333$.

Observation: SSE decreased from 7.5 → 5.8333 after recomputing centroids, as expected for K-means.

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4) Quick checklist for your own SSE computation

1. For each point x, compute squared distance to its cluster centroid: $||x - \mu_k||^2$.
2. Sum those squared distances for all points in a cluster → cluster SSE.
3. Sum cluster SSEs across all K clusters → total SSE.
4. When you reassign and recompute means, SSE should not increase (K-means decreases or keeps same).
